

Arduino Microcontroller Concepts (CSE 4820)

Introduction to Arduino Microcontroller



Program: B. TECH., 7th Semester

SIKSHA 'O' ANUSHANDHAN (Deemed to be) UNIVERSITY

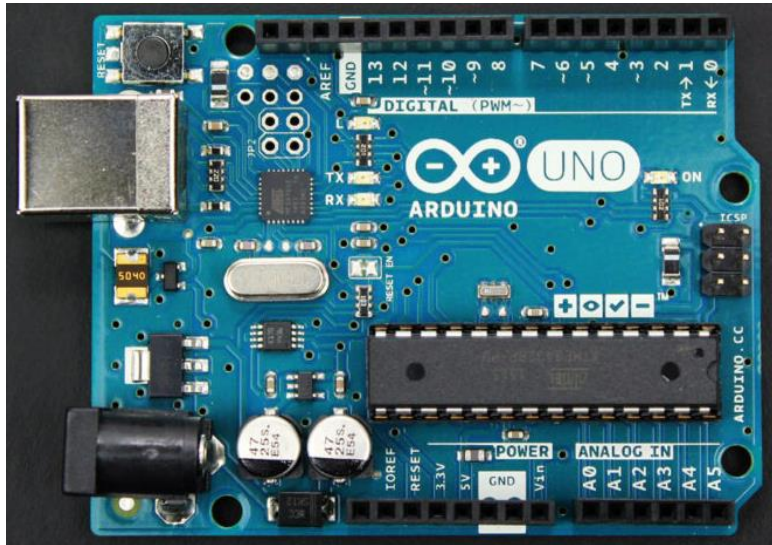
Arduino: The Brains of an Embedded System

‘What is Arduino?’

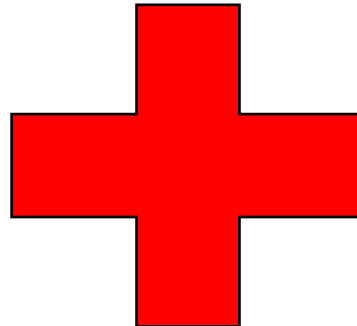
‘Why is it so popular?’

How does an Arduino board work?

Symbol of Arduino IDE



Hardware (physical programmable circuit board)



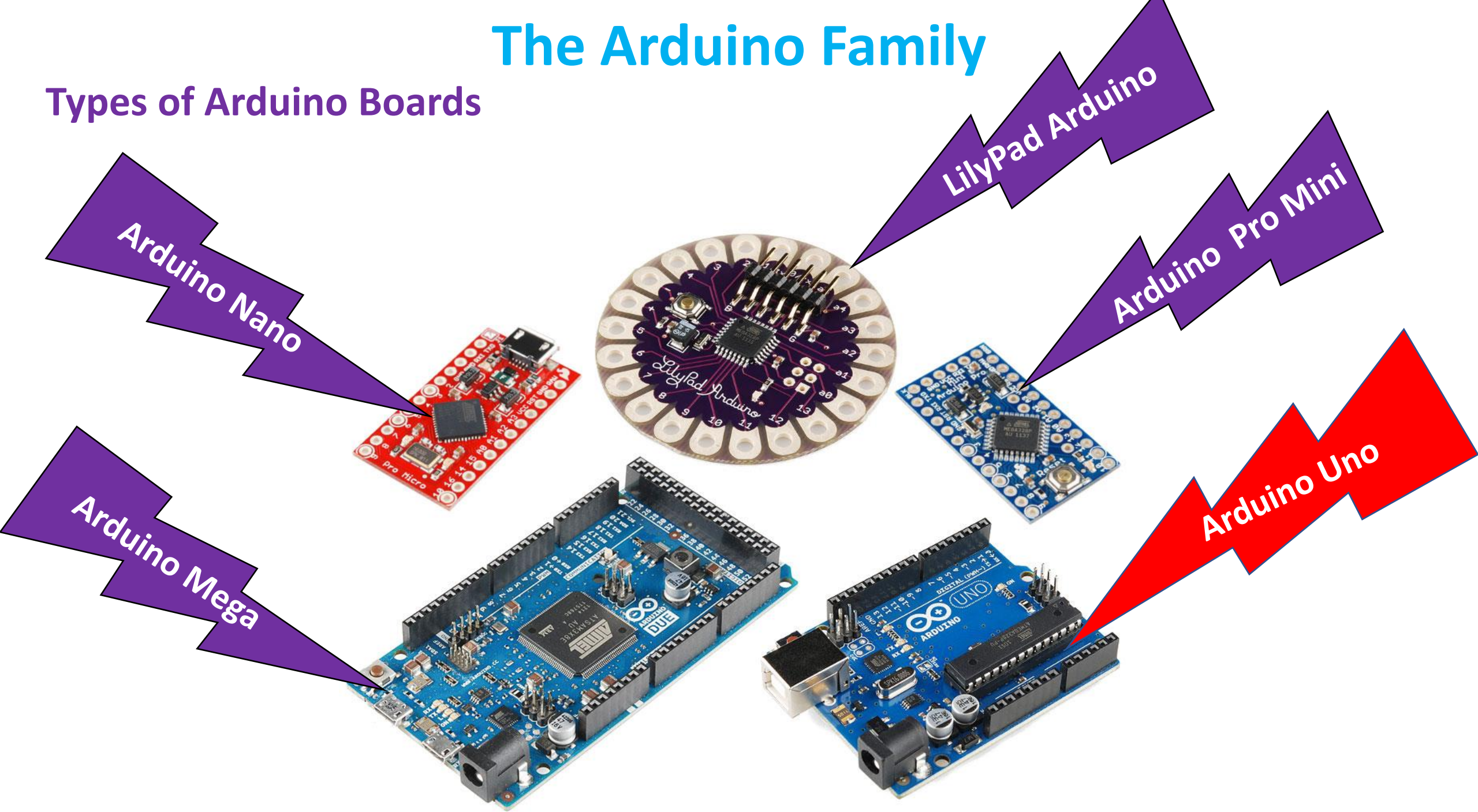
Software (Integrated Development Environment)

What Makes Arduino So Special?

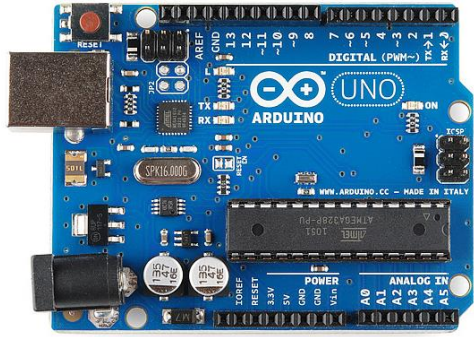
- Easy on your Wallet
- Operating system who?
- Easy to get started & massive following
- Open Source & Beautiful

The Arduino Family

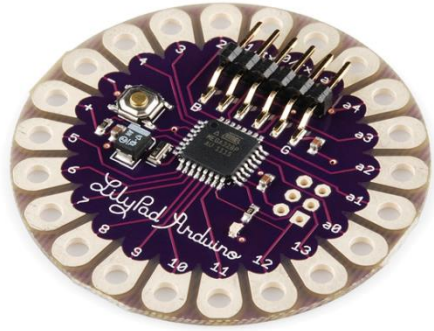
Types of Arduino Boards



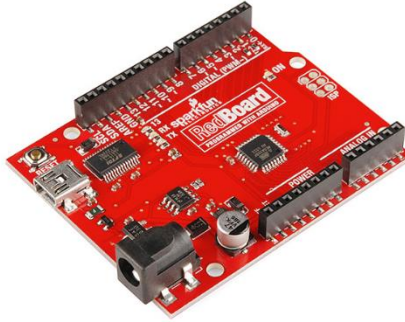
The Arduino Family contd.



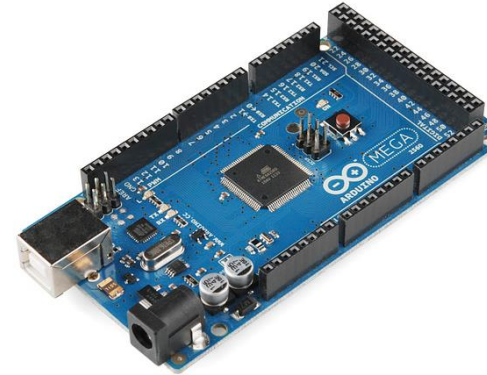
Arduino Uno (R3)



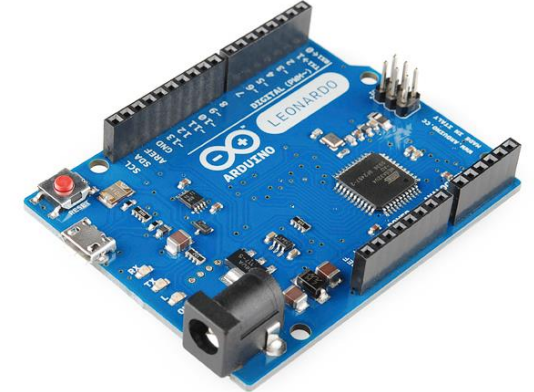
LilyPad Arduino



RedBoard

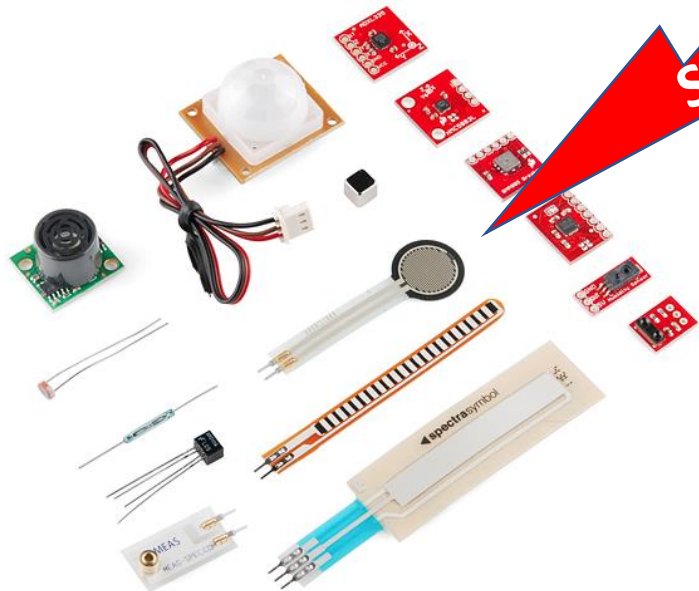


Arduino Mega (R3)

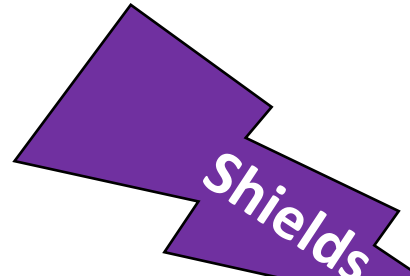


Arduino Leonardo

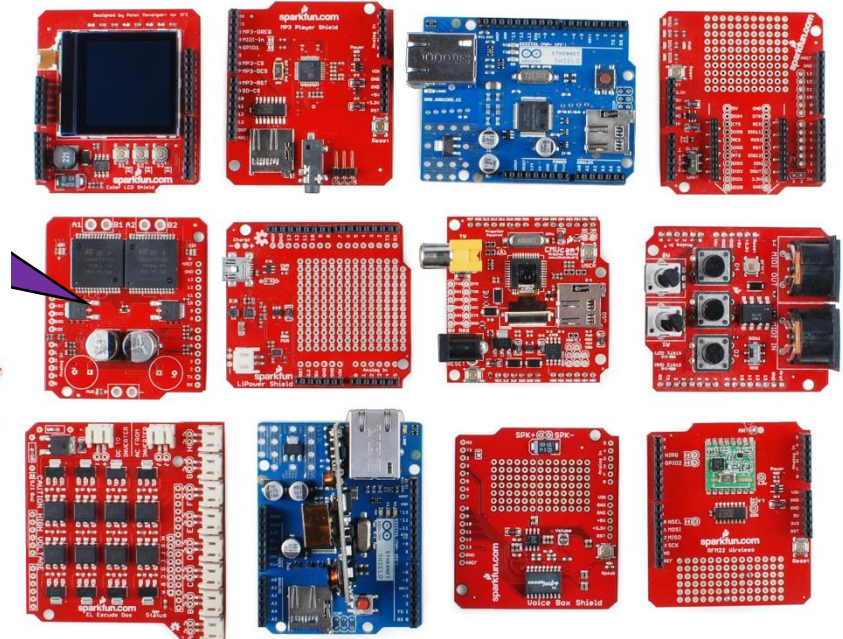
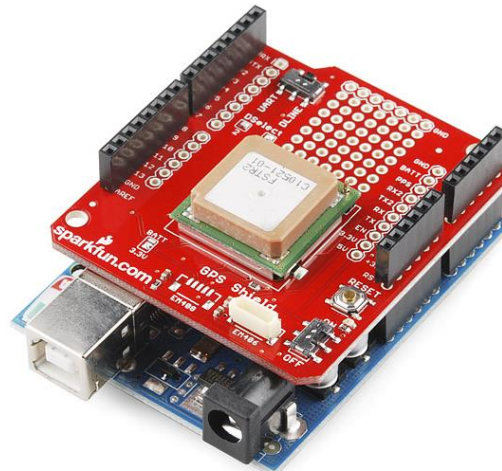
The Extended Family



Sensors



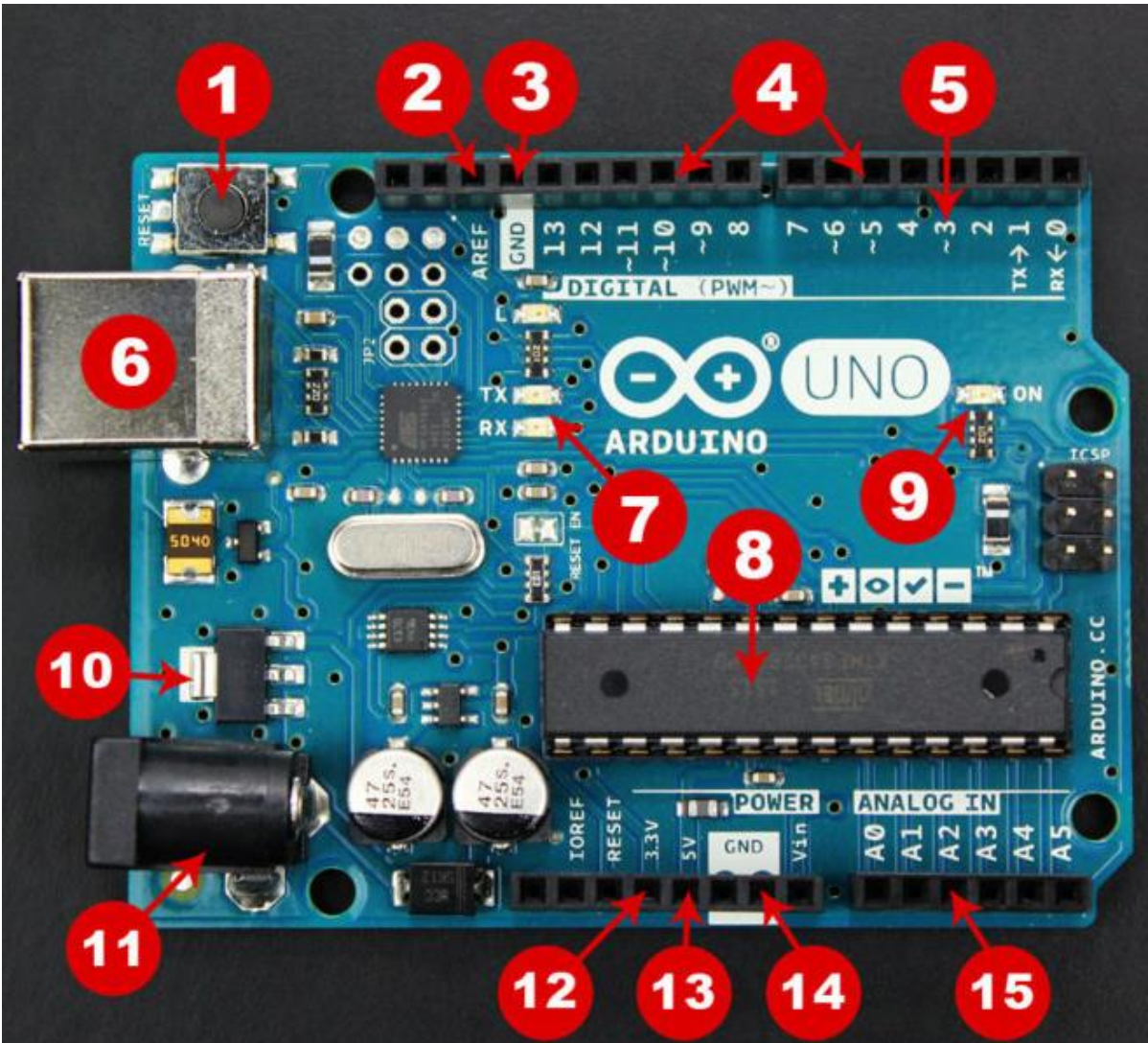
Shields



Arduino Uno

Board Breakdown

- | | |
|---------------------------|--------------------------|
| 1) Reset Button | 9) Power LED Indicator |
| 2) AREF | 10) Voltage Regulator |
| 3) Ground Pin | 11) DC Power Barrel Jack |
| 4) Digital Input/Output | 12) 3.3V Pin |
| 5) PWM | 13) 5V Pin |
| 6) USB Connection | 14) Ground Pins |
| 7) TX/RX | 15) Analog Pins |
| 8) ATmega Microcontroller | |



Arduino UNO Contd.

Basics of Arduino UNO

- Microcontroller board designed by **Arduino.cc**
- Atmega328 Microcontroller
- open-source platform
- Easy Libraries Design

Arduino UNO Pinout

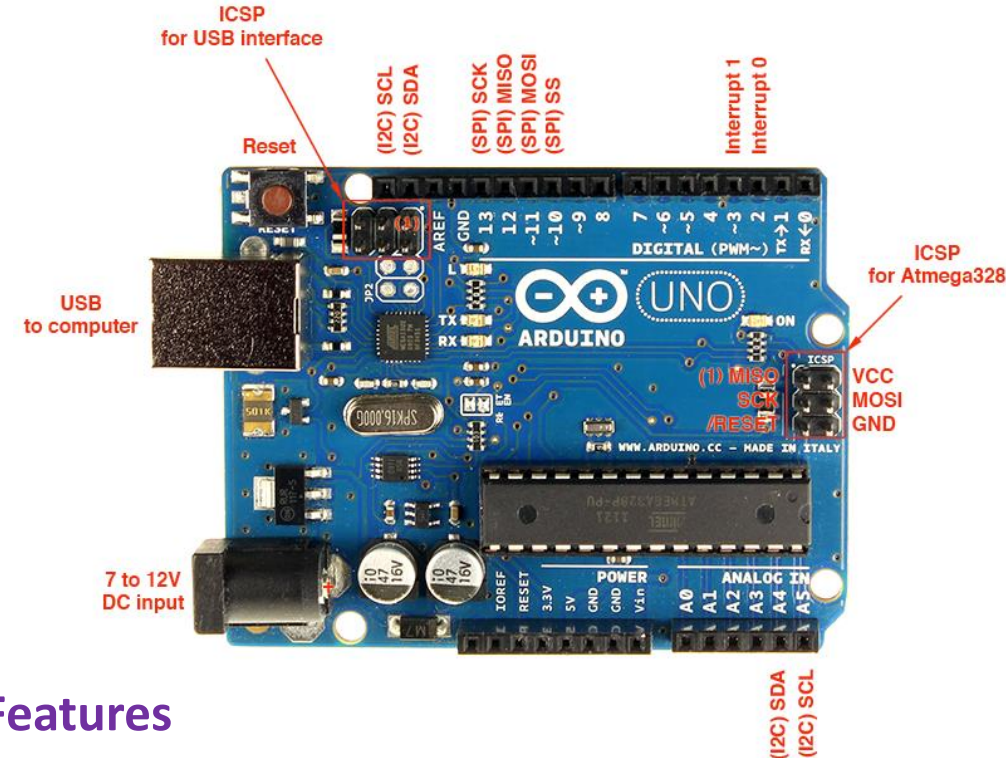
- 20 input/output pins (14 digital pins (including 6 PWM pins) + 6 analogue pins))
- 3 communication protocols:
 - ✓ Serial Protocol
 - ✓ I2C Protocol
 - ✓ SPI Protocol

Arduino UNO Memory Features

- flash memory of 32Kb
- SRAM of 2KB
- EEPROM memory of 1Kb
- Bootloader of 2Kb is installed so we are left with 30kb Flash memory.

Arduino UNO Applications

- Embedded Systems
- Control Systems
- Robotics
- Instrumentation
- Condition Monitoring



Other Features

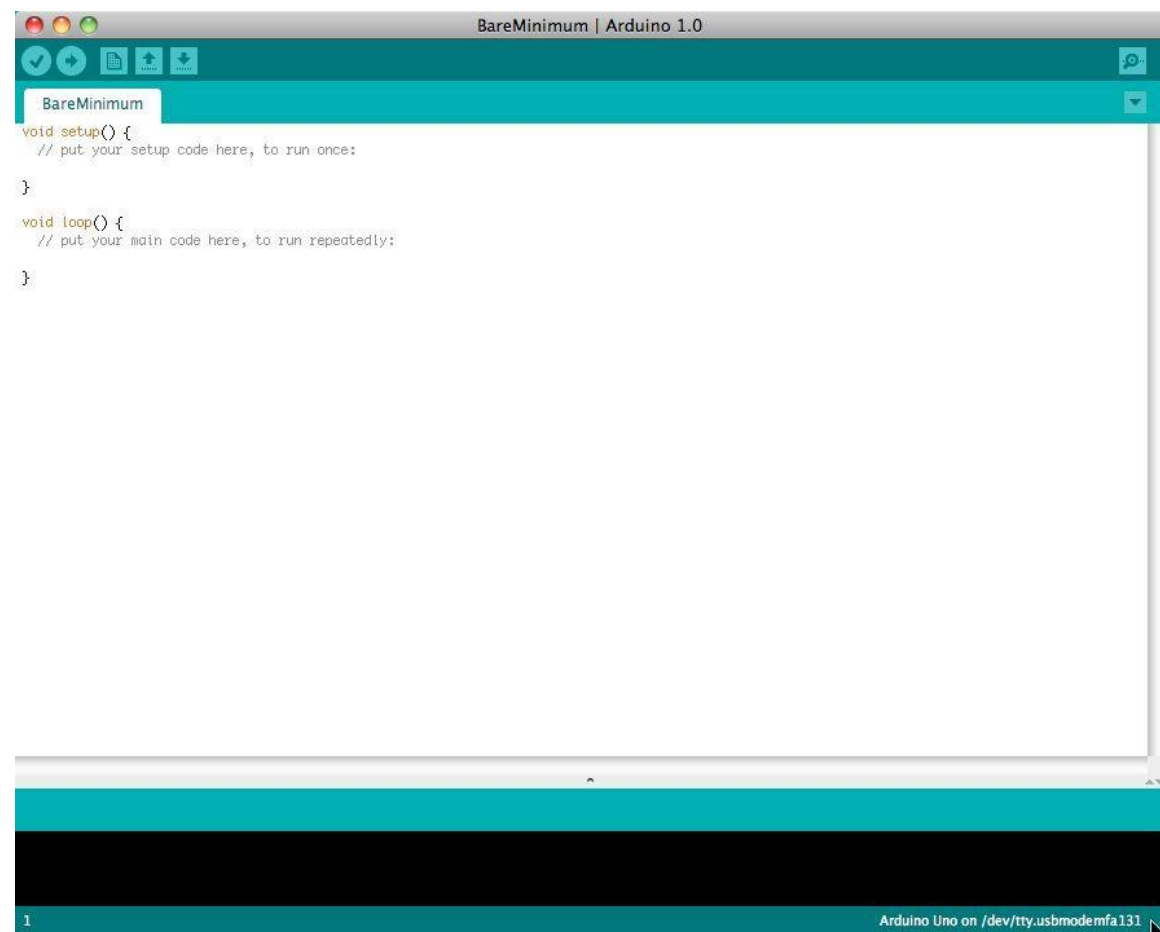
Operating Voltage	5V
Input Voltage (recommended)	7-12V
Input Voltage (limit)	6-20V
DC Current per I/O Pin	20 mA
DC Current for 3.3V Pin	50 mA
Clock Speed	16 MHz
LED_BUILTIN	13
Length : Width : Weight	68.6mm : 53.4mm : 25gm

How to get started with ARDUINO CODE

Arduino Coding Environment and basic tools

What language is Arduino?

Arduino IDE



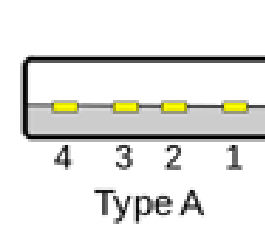
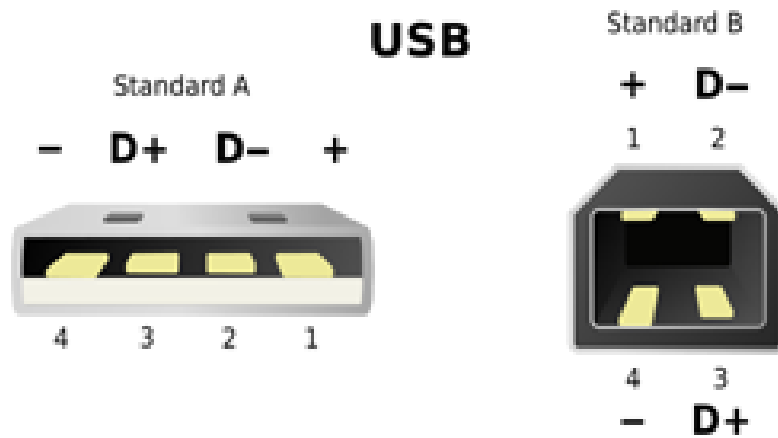
Installing Arduino IDE



USB Guide

What Kinds of Cables are Out There?

What Does USB Do?



USB-A



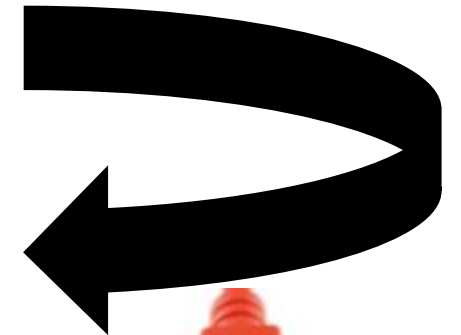
Micro-B



Mini-B



USB-B



- Stands for "Universal Serial Bus." **USB** is the most common type of computer port used in today's computers. It can be used to connect keyboards, mice, game controllers, printers, scanners, digital cameras, and removable media drives, just to name a few.
- **USB Types: Various Types of USB Cables (A, B and C) and Their Differences.** Based on physical design of the plugs and ports the three different **types** of USB **cable** are: USB **Type A**, USB **Type B** and USB **Type C**. Based on the functionality of USB connectors there are also two different versions of USB: USB 2.0 and USB 3.0.

What Kind of Cable Do I Need?

USB A to Mini-B



USB A to Micro-B



USB A to B



USB A to A



USB 2.0 type A
to Mini-B 5-pin cable



USB 2.0 type A
to micro USB 5-pin cable



standard issue USB 2.0 cable

USB Extension Cable

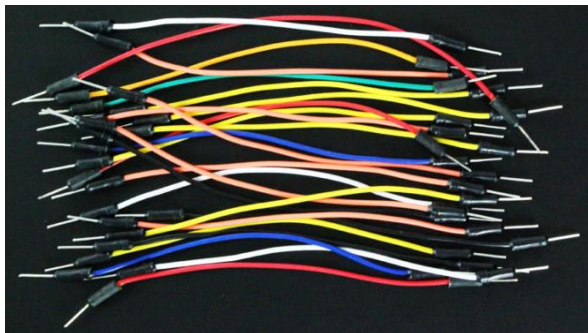
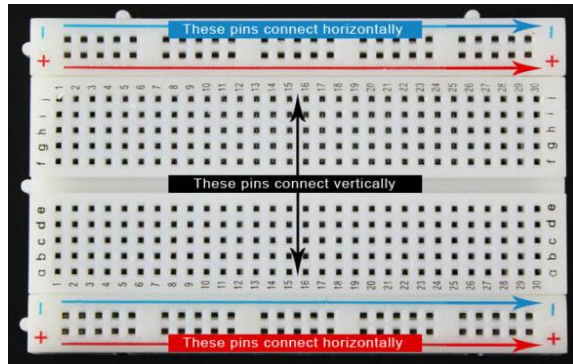


How an Arduino circuit looks?

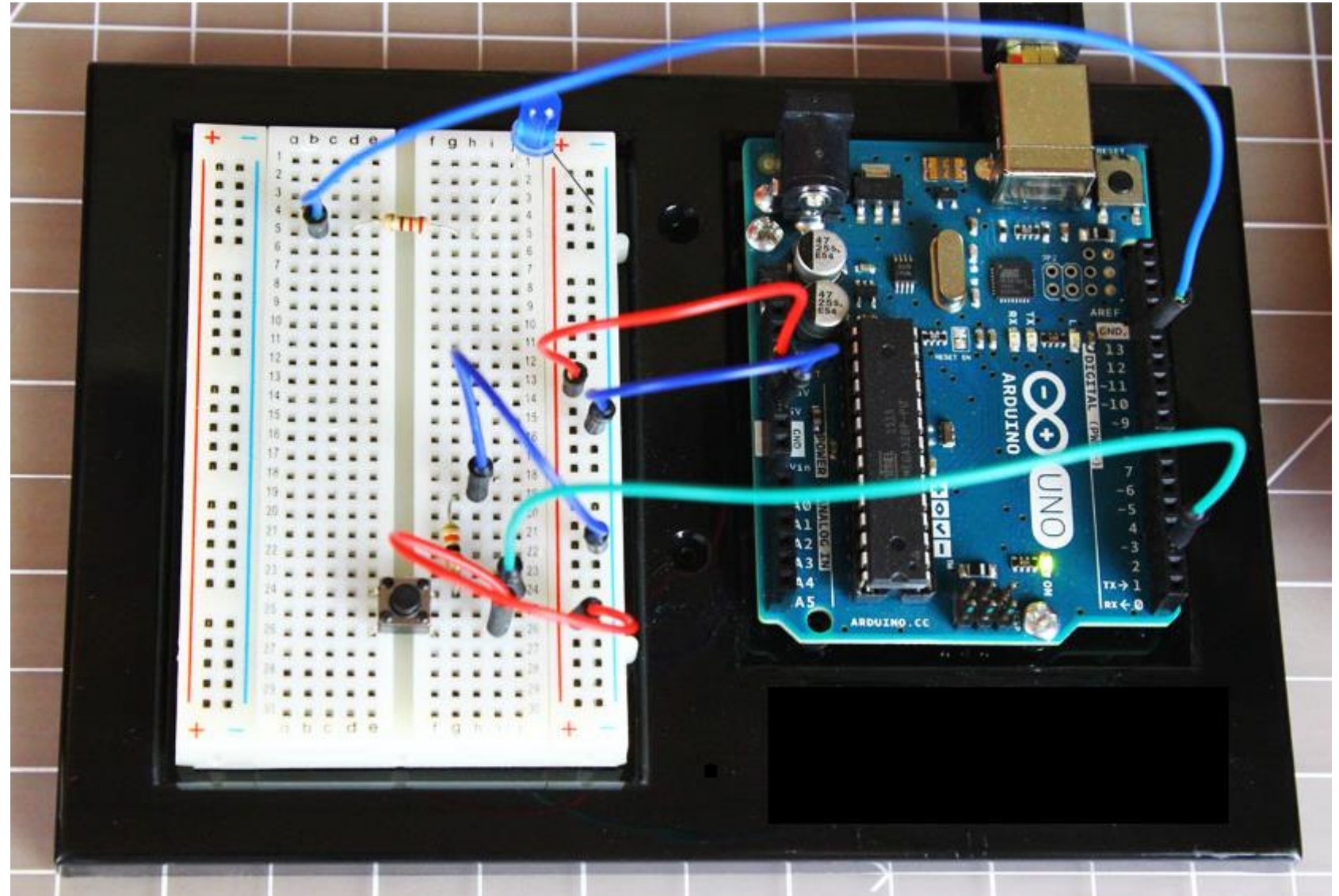
Arduino Power Supply



Arduino Breadboard



Completed Arduino Circuit



Installing Arduino IDE

Step 1 – First you must have your Arduino board (you can choose your favorite board) and a USB cable.



Arduino UNO, Arduino Duemilanove, Nano, Arduino Mega 2560, or Diecimila : A plug to B plug

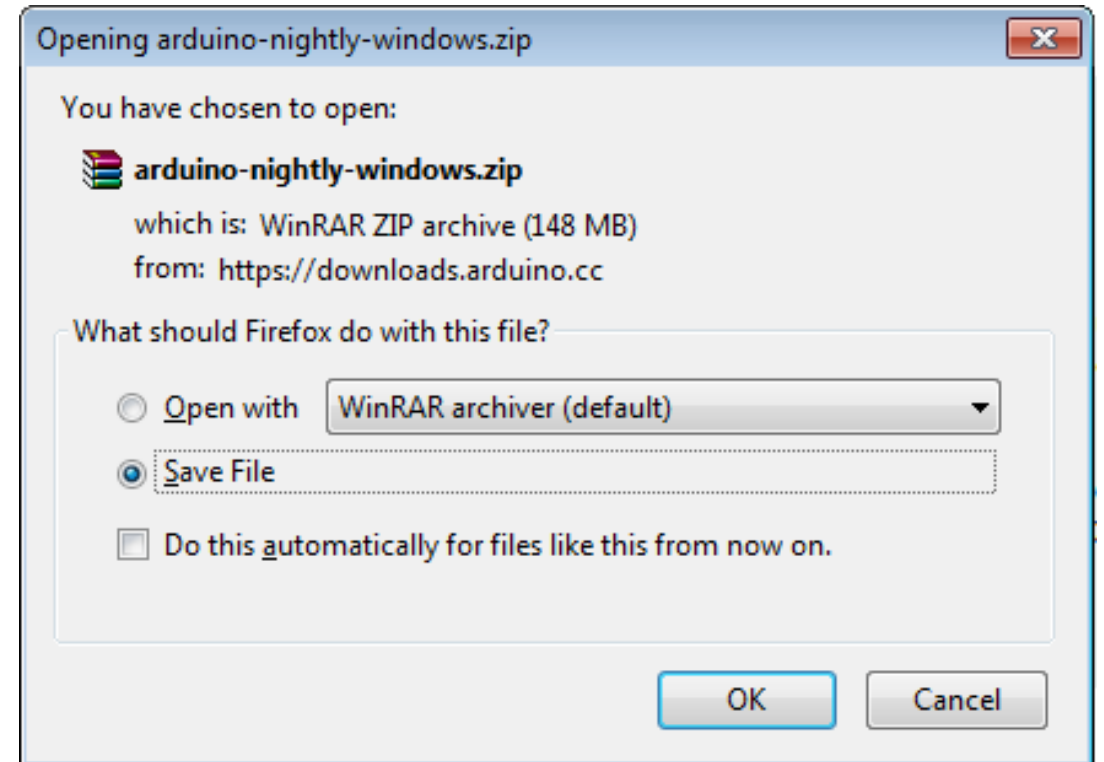


Arduino Nano :
A to Mini-B cable

Step 3 – Power up your board.

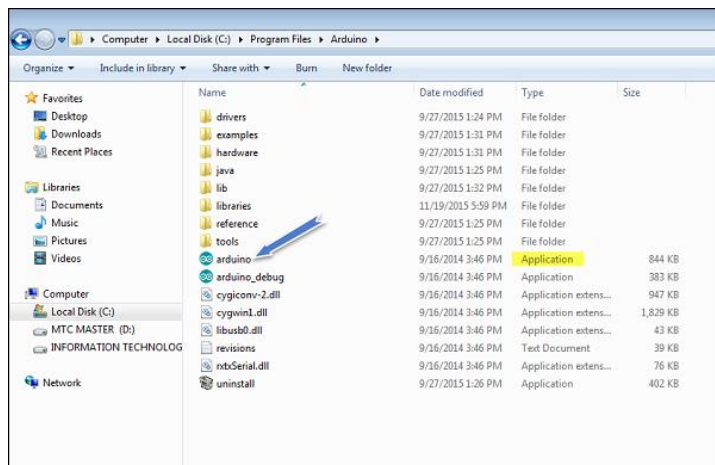
Connect the Arduino board to your computer using the USB cable. The green power LED (labeled PWR) should glow.

Step 2 – Download Arduino IDE Software.



Installing Arduino IDE contd.

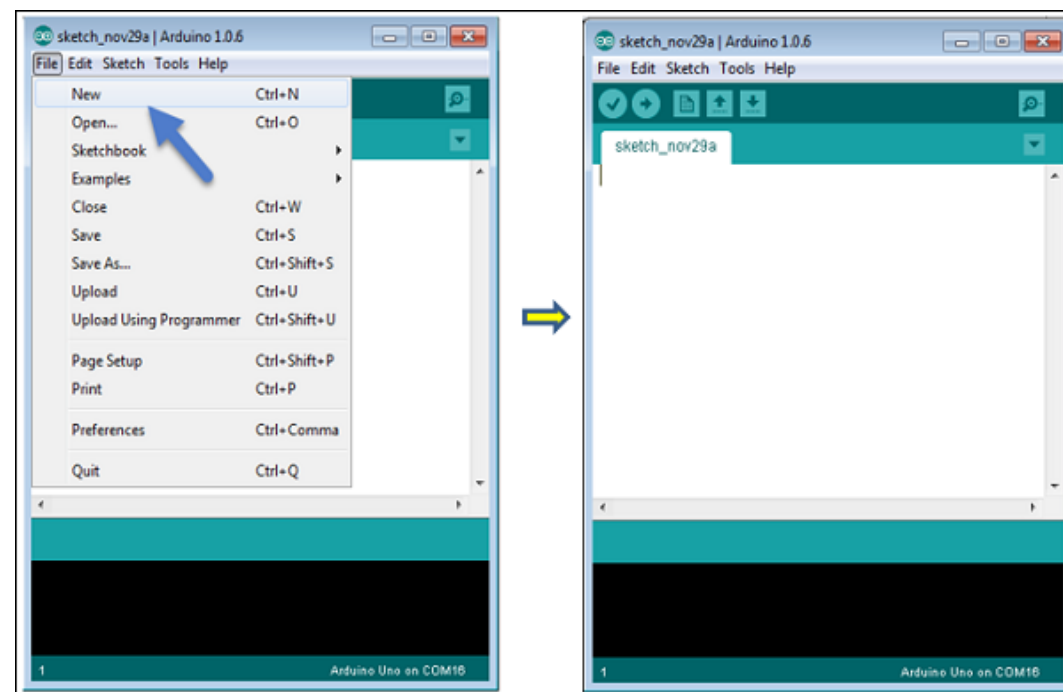
Step 4 – Launch Arduino IDE.



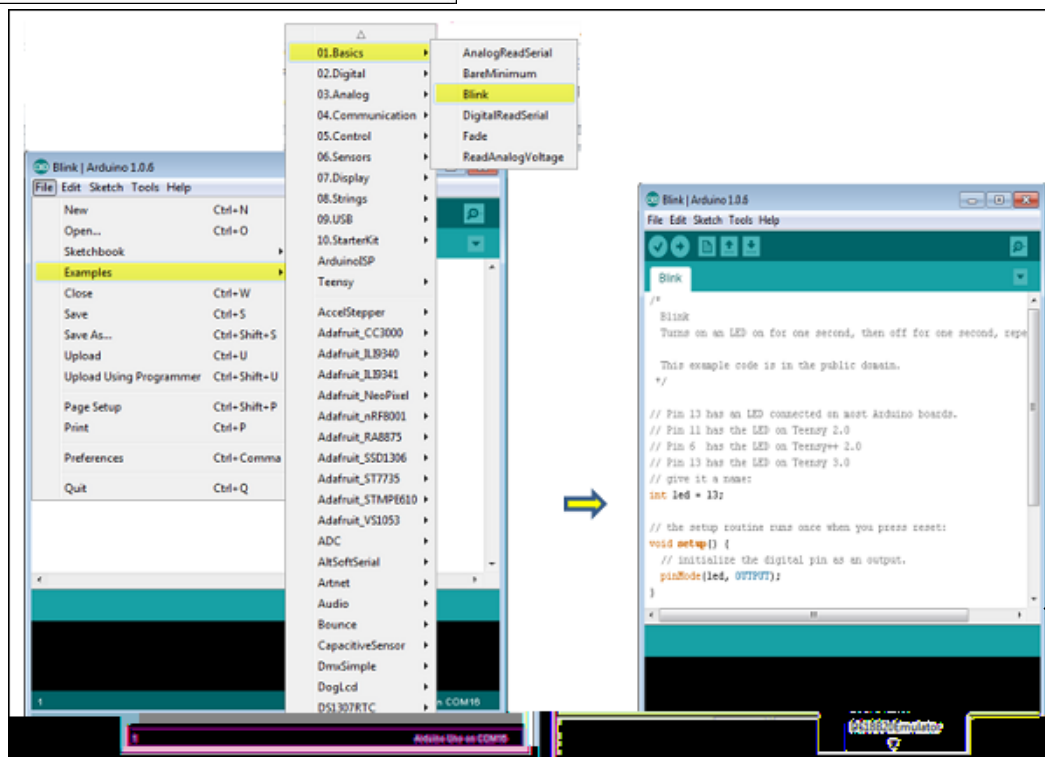
Step 5 – Open your first project.

- Create a new project.
- Open an existing project example.

To create a new project, select File → New.



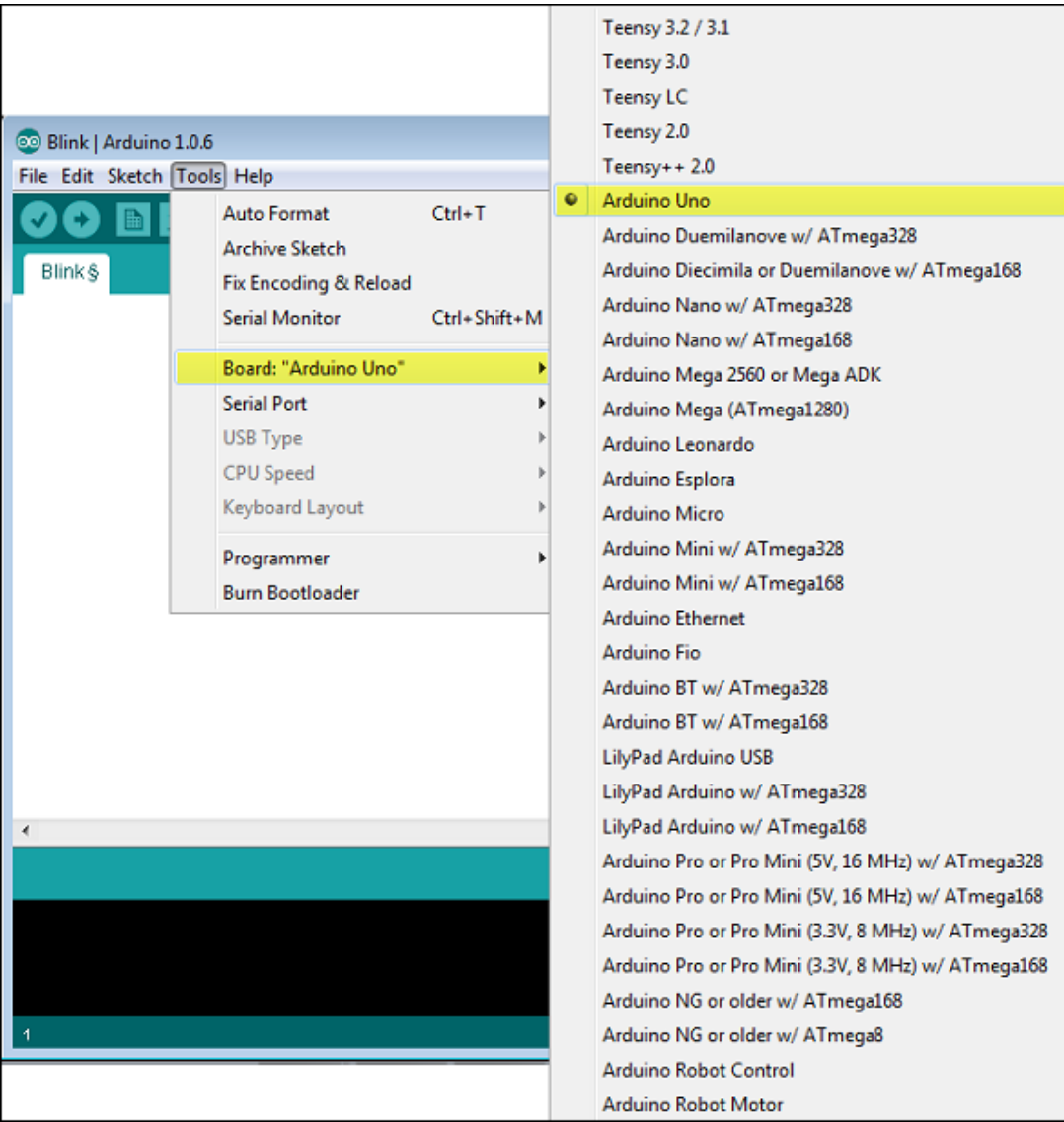
To open an existing project example, select File → Example → Basics → Blink.



Installing Arduino IDE contd.

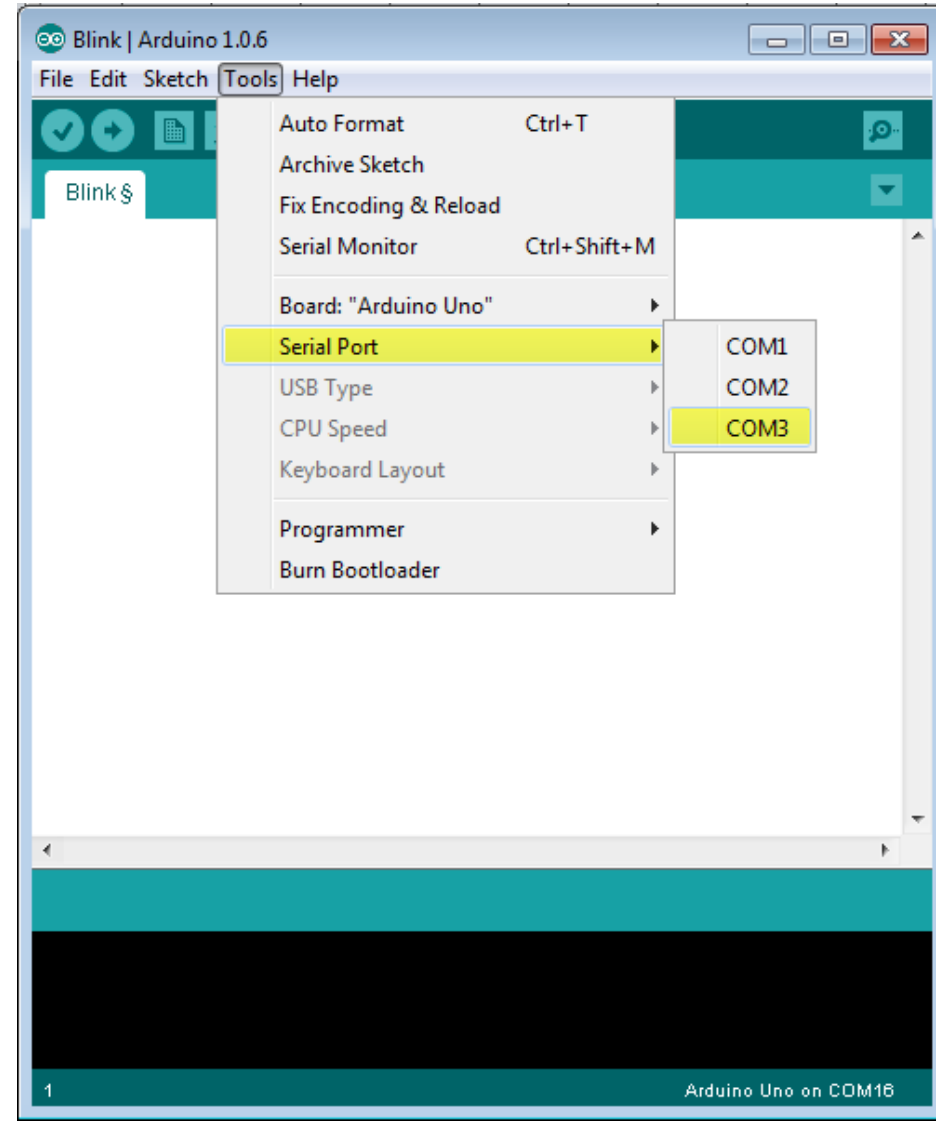
Step 6 – Select your Arduino board.

Go to Tools → Board and select your board.



Step 7 – Select your serial port.

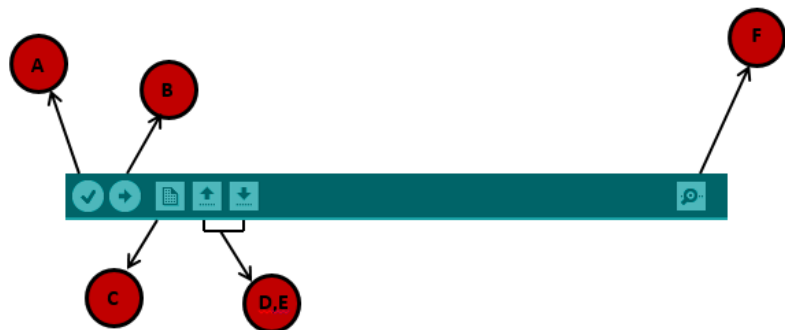
Go to Tools → Serial Port menu



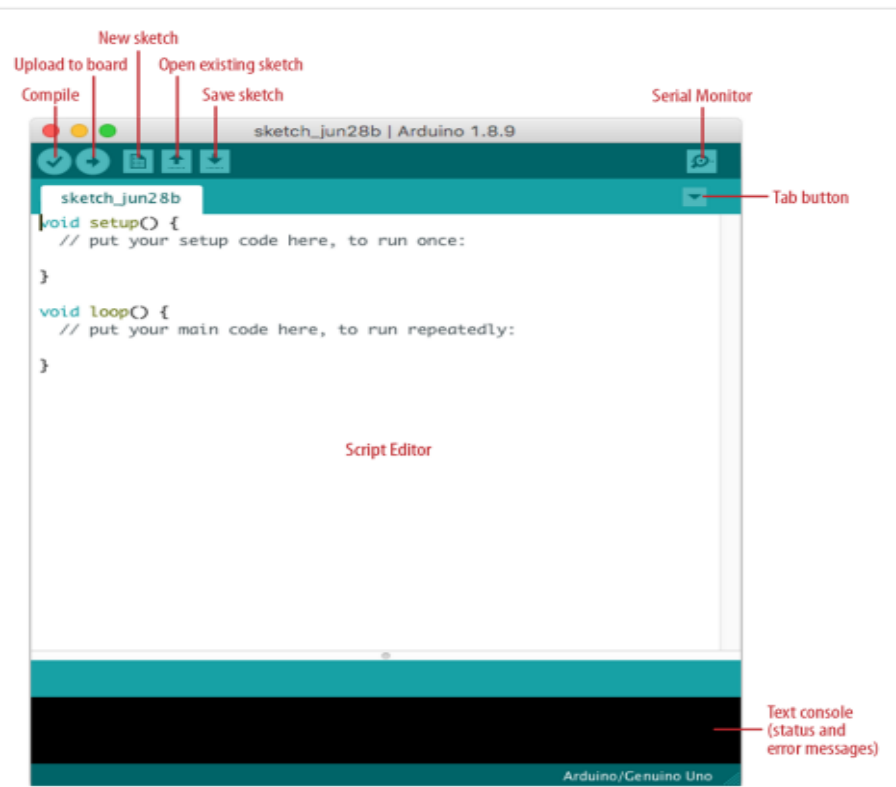
COM3 or higher

Installing Arduino IDE contd.

Step 8 – Upload the program to your board.



- A** – Used to check if there is any compilation error.
- B** – Used to upload a program to the Arduino board.
- C** – Shortcut used to create a new sketch.
- D** – Used to directly open one of the example sketch.
- E** – Used to save your sketch.
- F** – Serial monitor used to receive serial data from the board and send the serial data to the board.



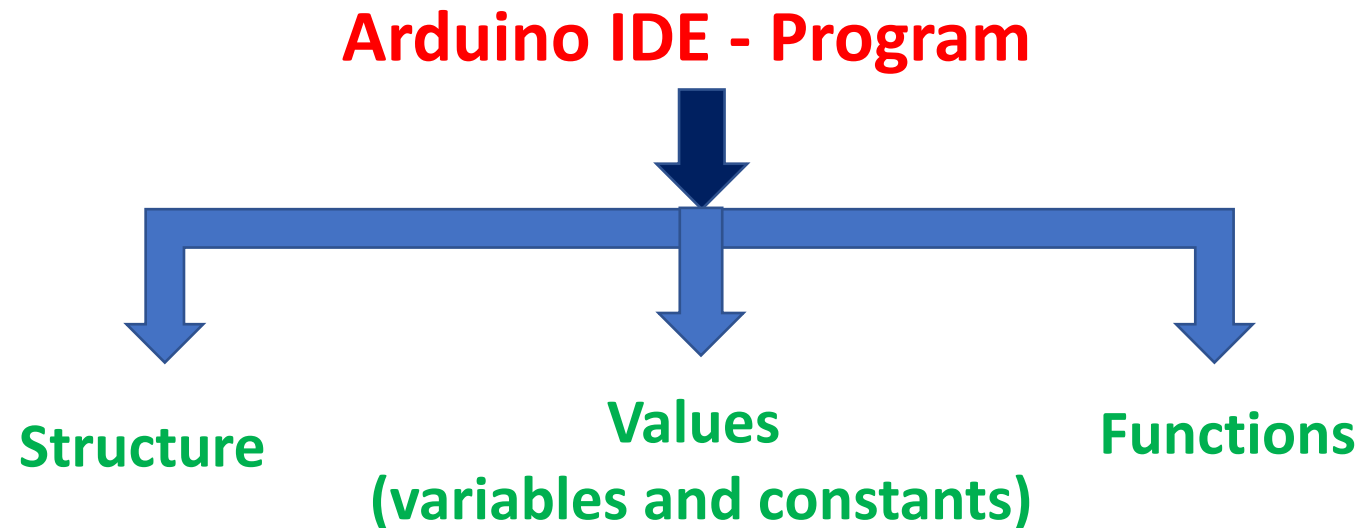
Note – If you have an Arduino Mini, NG, or other board, you need to press the reset button physically on the board, immediately before clicking the upload button on the Arduino Software.

Arduino IDE - Program Structure

The source code for the Java environment : **GPL**

The C/C++ microcontroller libraries : **LGPL**

Arduino program : **Sketch**



Structure

Software structure consist of two main functions –

Setup() function

Loop() function

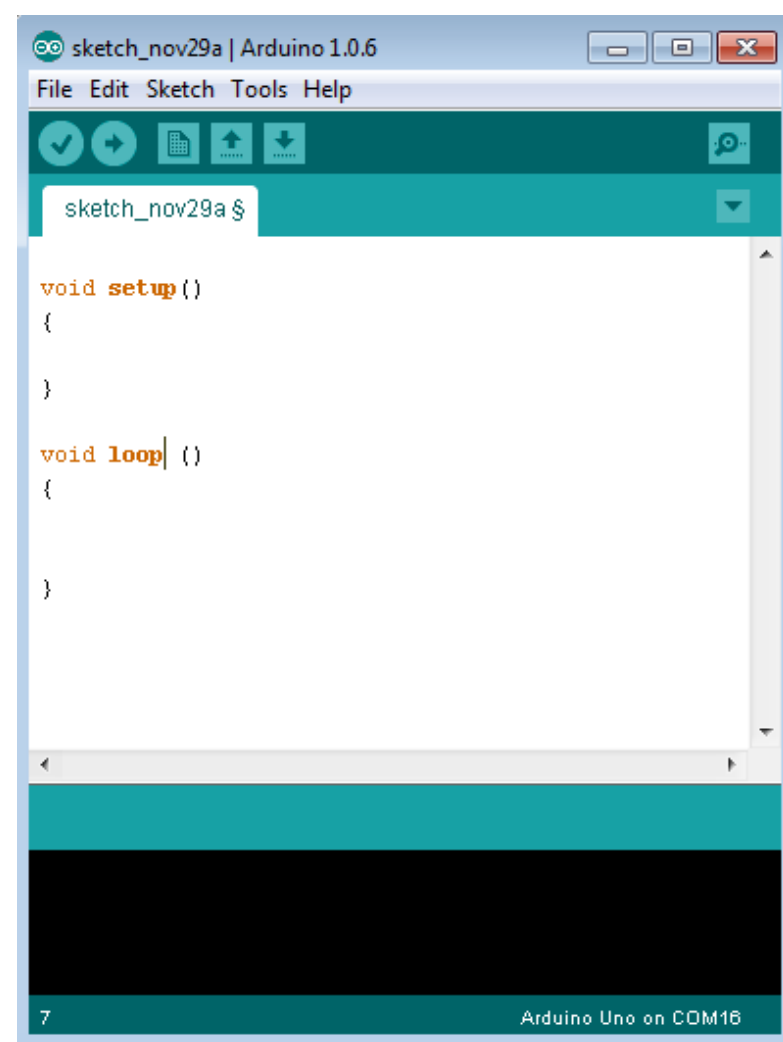
```
Void setup ( ) {  
}
```

PURPOSE – Sets things up that have to be done once and then don't happen again.

```
Void loop ( ) {  
}
```

PURPOSE – Contains the instructions that get repeated over and over until the board is turned off.

```
void setup()  
{  
  pinMode(LED_BUILTIN, OUTPUT);  
}  
  
void loop()  
{  
  digitalWrite(LED_BUILTIN, HIGH); // set the LED on  
  delay(2000);                     // wait for two seconds  
  digitalWrite(LED_BUILTIN, LOW);  // set the LED off  
  delay(2000);                     // wait for two seconds  
}
```



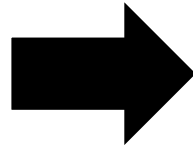
Arduino - I/O Functions

pinMode() Function

- ✓ used to configure a specific pin to behave either as an input or an output.

pinMode() Function Syntax

```
void setup () {  
  pinMode (pin , mode);  
}
```



pin – the number of the pin whose mode you wish to set

mode – INPUT, OUTPUT, or INPUT_PULLUP.

Pins Configured as INPUT

- ✓ Arduino pins are by default configured as inputs

Input → pinMode() → high-impedance state

i.e., it takes very little current to switch the input pin from one state to another.

Pins configuration → pinMode(pin, INPUT)

Pull-up Resistors

- ✓ useful to steer an input pin to a known state if no input is present.
- ✓ built-in pull-up resistors are accessed by setting the pinMode() as **INPUT_PULLUP** which effectively **inverts** the behavior of the **INPUT** mode, where HIGH means the sensor is OFF and LOW means the sensor is ON.

```
pinMode(3,INPUT);  
pinMode(5,INPUT_PULLUP);
```

```
// set pin to input without using built in pull up resistor  
// set pin to input using built in pull up resistor
```

Arduino - I/O Functions contd.

Pins Configured as OUTPUT

Output  **pinMode()**  **Low-impedance state**

i.e., they can provide a substantial amount of current (upto 40 mA) to other circuits.

Attempting to run high current devices from the output pins, can damage or destroy the output transistors in the pin, or damage the entire Atmega chip. Often, this results in a "dead" pin in the microcontroller but the remaining chips still function adequately.

Pins configuration  **pinMode(pin, OUTPUT)**

```
int button = 5 ;
int LED = 6;
```

```
// button connected to pin 5
// LED connected to pin 6
```

```
void setup () {
  pinMode(button , INPUT_PULLUP);
  pinMode(button , OUTPUT);
}
```

```
// set the digital pin as input with pull-up resistor
// set the digital pin as output
```

```
void loop () {
  If (digitalRead(button ) == LOW)           // if button pressed {
    digitalWrite(LED,HIGH);                  // turn on led
    delay(500);                             // delay for 500 ms
    digitalWrite(LED,LOW);                   // turn off led
    delay(500);                             // delay for 500 ms
  }
}
```

Arduino - I/O Functions contd.

digitalWrite() Function

- ✓ used to write a HIGH or a LOW value to a digital pin.

If you do not set the `pinMode()` to `OUTPUT`, and connect an LED to a pin, when calling `digitalWrite(HIGH)`, the LED may appear dim. Without explicitly setting `pinMode()`, `digitalWrite()` will have enabled the internal pull-up resistor, which acts like a large current-limiting resistor.

digitalWrite() Function Syntax

```
void loop() {  
    digitalWrite (pin ,value);  
}
```

pin – the number of the pin whose mode you wish to set

value – HIGH, or LOW.

```
int LED = 6;           // LED connected to pin 6
```

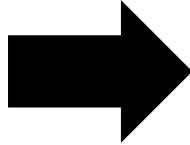
```
void setup () {  
  pinMode(LED, OUTPUT); // set the digital pin as output  
}
```

```
void loop () {  
    digitalWrite(LED,HIGH);    // turn on led  
    delay(500);                // delay for 500 ms  
    digitalWrite(LED,LOW);     // turn off led  
    delay(500);                // delay for 500 ms  
}
```

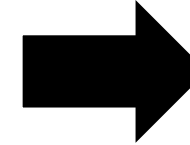
Arduino - I/O Functions contd.

analogRead() function

digitalRead() function

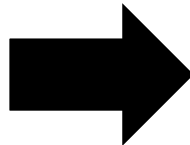


Arduino is able to detect whether there is a voltage applied to one of its pins

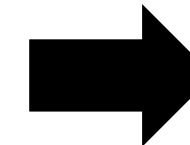


on/off sensor

analogRead() function



Arduino can read the voltage applied to one of the "Analog In" pins.



analog sensor, whose value continuously changes

analogRead() function returns a number between 0 and 1023, which represents voltages between 0 and 5 volts.

analogRead() function Syntax

analogRead(pin);

pin – the number of the analog input pin to read from (0 to 5 on most boards, 0 to 7 on the Mini and Nano, 0 to 15 on the Mega)

```
int analogPin = 3;  
int val = 0;
```

```
//potentiometer wiper (middle terminal) connected to analog pin 3  
// variable to store the value read
```

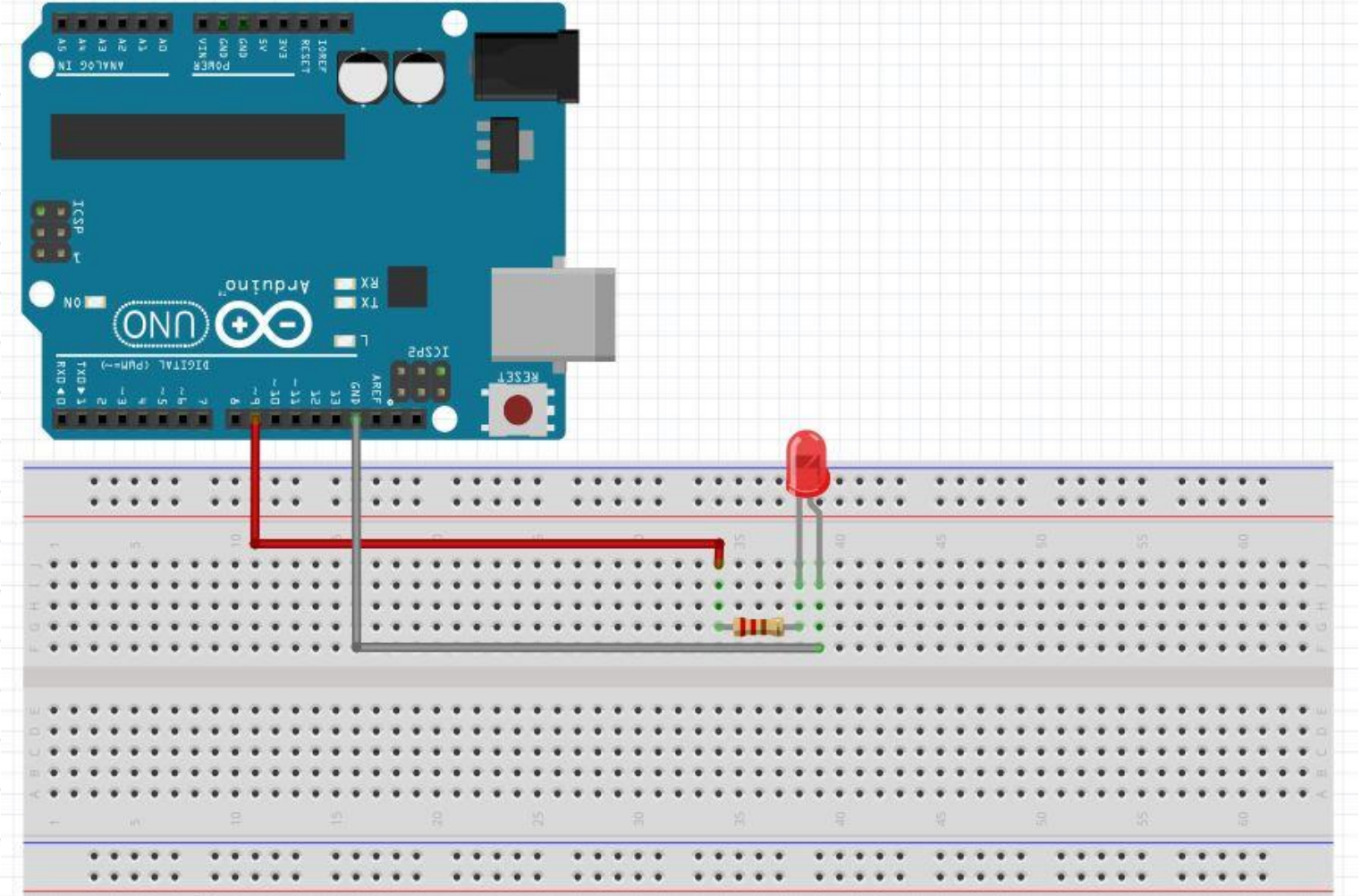
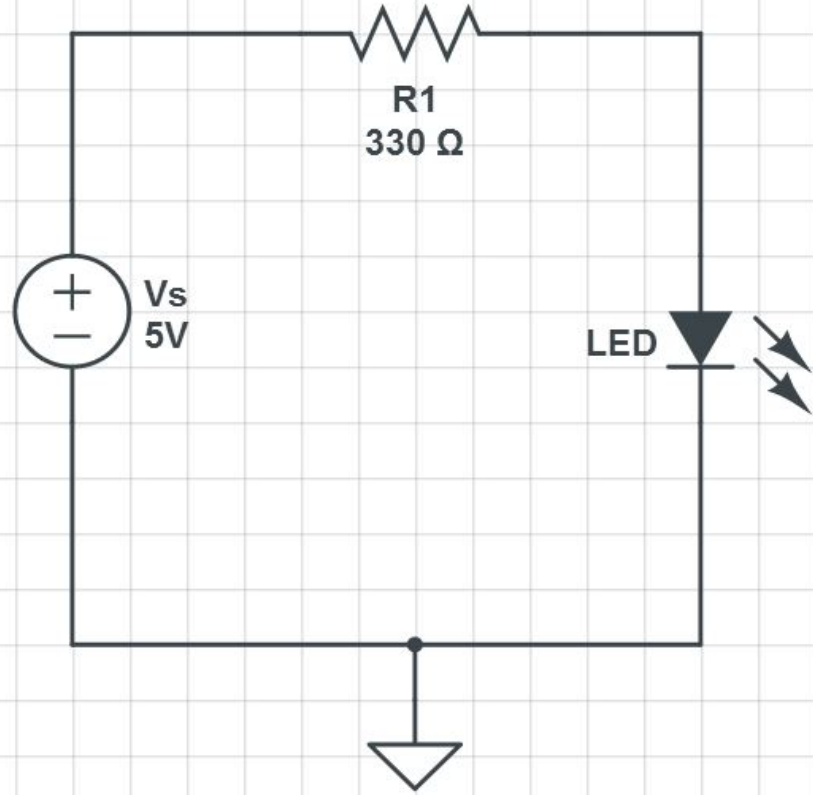
```
void setup() {  
  Serial.begin(9600);  
}
```

```
// setup serial
```

```
void loop() {  
  val = analogRead(analogPin);  
  Serial.println(val);  
}
```

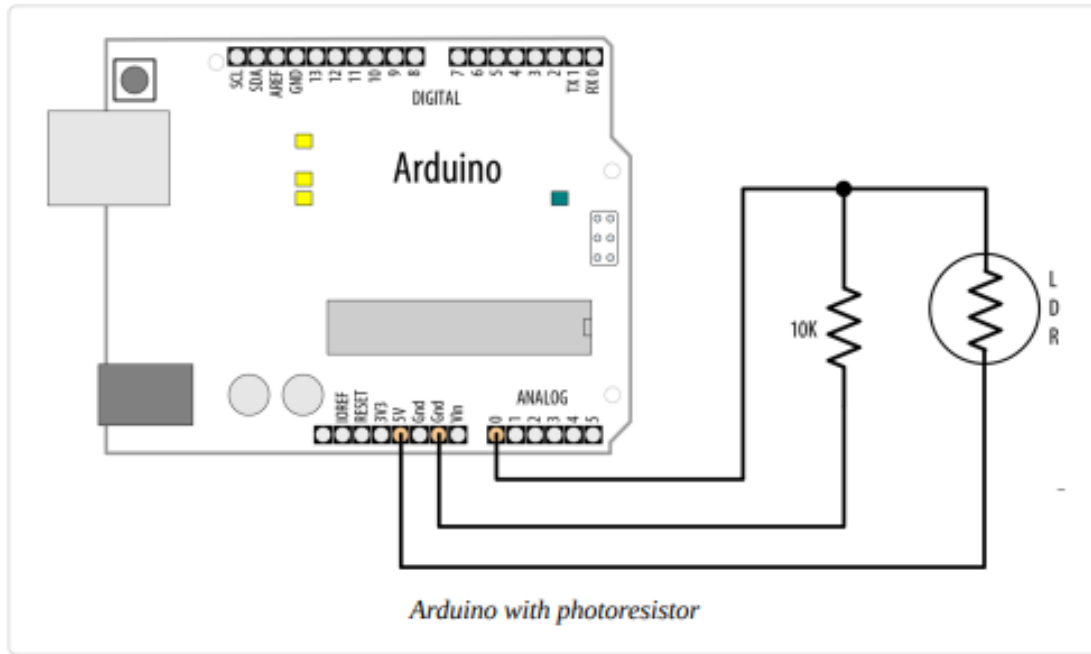
```
// read the input pin  
// debug value
```

Arduino – Write your own code



An Easy Arduino Project

(Arduino with Photoresistor)



The following sketch reads the light level of a photoresistor connected to analog pin 0. The light level striking the photoresistor will change the blink delay of the internal onboard LED:

```
/*
 * Blink with photoresistor sketch
 */
const int sensorPin = A0;           // connect sensor to analog input 0

void setup()
{
  pinMode(LED_BUILTIN, OUTPUT); // enable output on the led pin
}

void loop()
{
  int delayval = analogRead(sensorPin); // read the analog input
  digitalWrite(LED_BUILTIN, HIGH);      // set the LED on
  delay(delayval);                      // delay is dependent on light level
  digitalWrite(LED_BUILTIN, LOW);       // set the LED off
  delay(delayval);
}
```

2 Mark Questions and Answers

1. What is Arduino and an Arduino sketch?

Arduino is an open-source electronics platform consisting of **hardware and software**. It can be used to build interactive electronic projects. An Arduino sketch is a **program written for an Arduino board**. It contains instructions that control the operation of the board.

2. What is Arduino IDE? Name the microcontroller used in Arduino Uno.

Arduino IDE is the software environment used to **write, edit, compile, and upload sketches** to an Arduino board. The Arduino Uno uses the **ATmega328P microcontroller**.

3. What is the purpose of the setup() and loop() function?

The setup() function contains instructions that are executed **once when the Arduino starts or resets**.

The loop() function contains instructions that are **repeated continuously** while the Arduino is powered.

4. What is the purpose of pinMode()? Explain with an example.

pinMode() is used to **configure an Arduino pin as an input or output**.

```
pinMode(LED_BUILTIN, OUTPUT);
```

5. What is the purpose of digitalWrite() and delay()?

digitalWrite() is used to set a digital output pin to **HIGH or LOW**.

```
digitalWrite(LED_BUILTIN, HIGH);
```

The delay() function **pauses the execution of the program for a specified time**.

```
delay(1000);
```

 Here, the delay is **1000 milliseconds (1 second)**.

6. Write the basic structure of an Arduino sketch.

```
void setup()
{
  // Initialization
}

void loop()
{
  // Repeated instructions
}
```

The setup() function executes once, while loop() executes repeatedly.

7. What is the purpose of the Blink program?

The Blink program demonstrates how to **turn the Arduino's built-in LED ON and OFF repeatedly**. The example uses setup() and loop() to control the LED.

8. What are the two basic steps involved in transferring a program to an Arduino board?

- The **two basic steps** involved in transferring a program to an Arduino board are:

Compile the program – The Arduino IDE converts the sketch into instructions that the Arduino can understand.

Upload the program – The compiled program is transferred from the computer to the Arduino board.